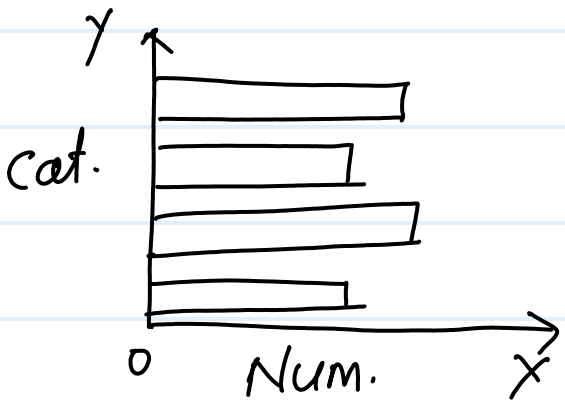
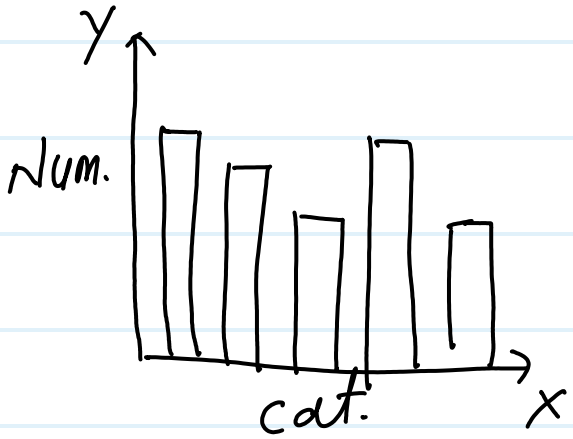
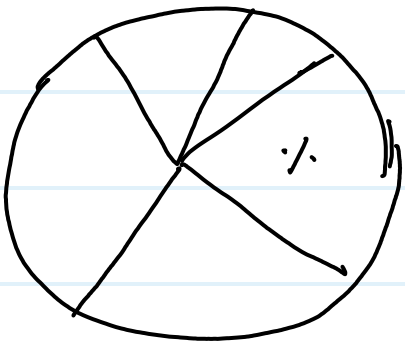
visualization

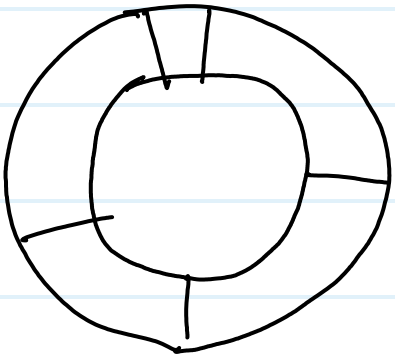
1) Bar chart & Colorum chart



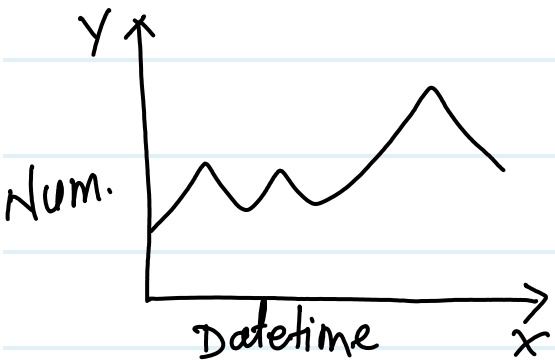


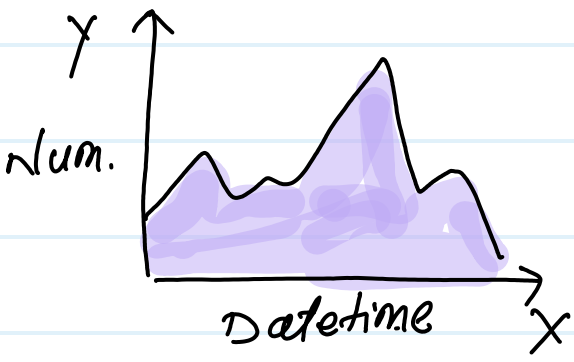
2) Pie chart & Donut chart



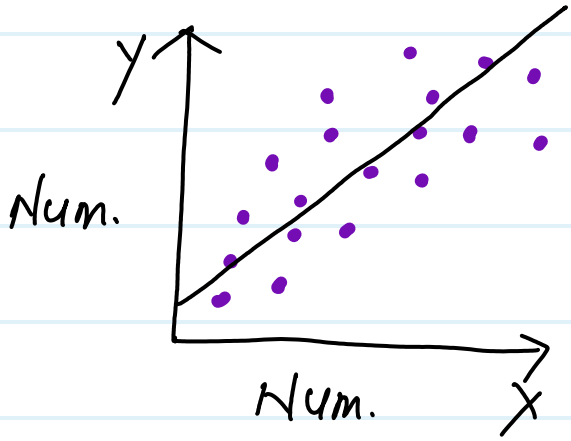


3) Line chart & Area chart

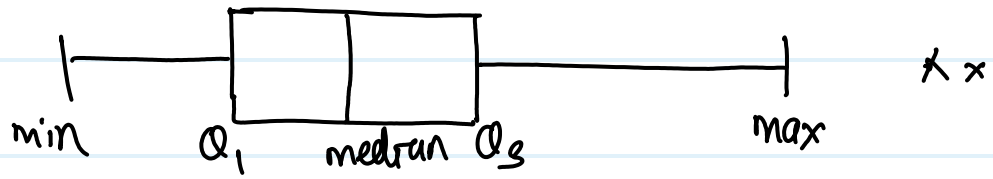




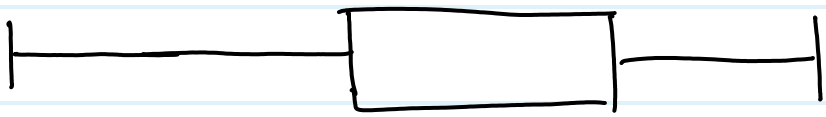
4) Scatter plot



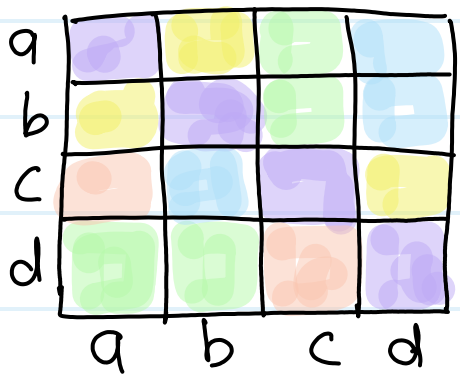
5) Box and whisker.



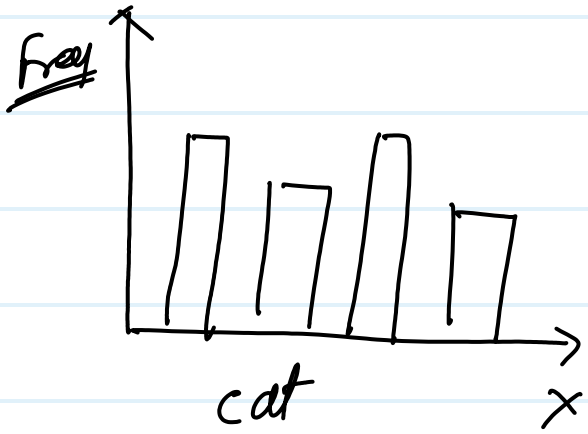
skewness



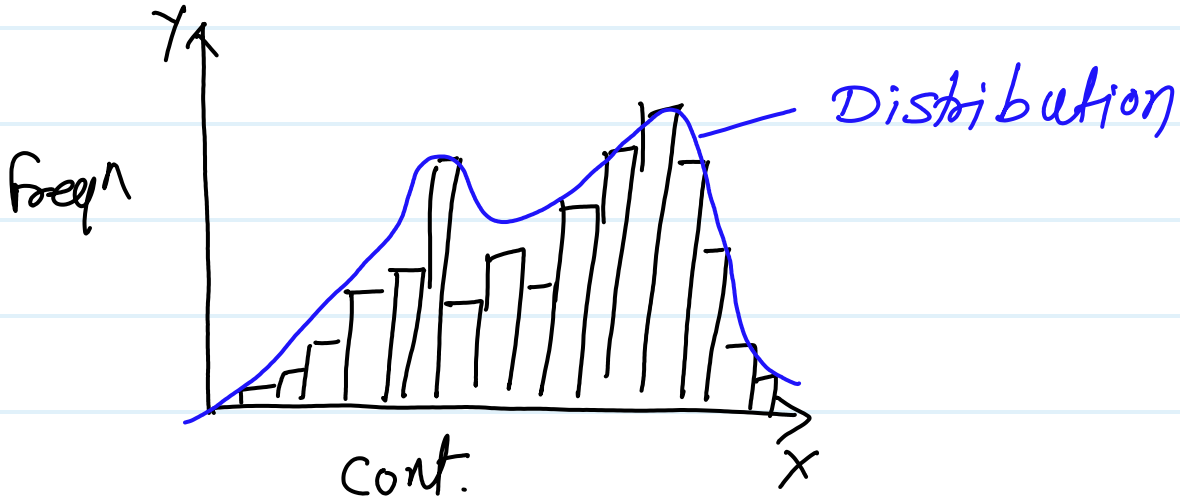
6) Heatmap



7) count plot



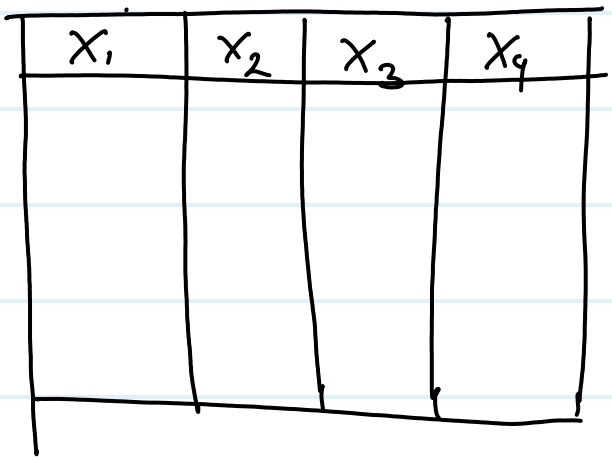
8) Histogram

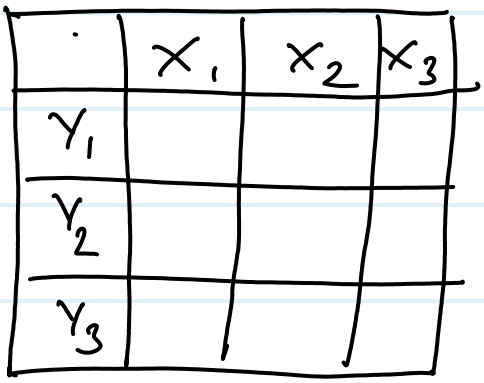


No of Bins –

Size of Bins –

9) Table & matrix





covariance

Age	height
10	100
12	120
14	140
16	160
18	170
20	180

$X\uparrow \quad Y\uparrow$ + covariance

$X\downarrow \quad Y\downarrow$ + covariance

$X\uparrow \quad Y\downarrow$ − covariance

$X\downarrow \quad Y\uparrow$ − covariance

$X\uparrow \quad Y$ zero covariance

$X\downarrow \quad Y$ zero covariance

$$cov(x,y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

Range $-\infty$ $+\infty$

x_1 and y = 415

x_2 and y = 50

x_3 and y = 10

x_4 and y = -40

x_5 and y = 900

co-relation

① pearson's corelation coefficient

$$P_{cov(x,y)} = \frac{cov(x,y)}{\sigma_x \sigma_y}$$

Range (−1 to +1)

x_1 and y = 0.8

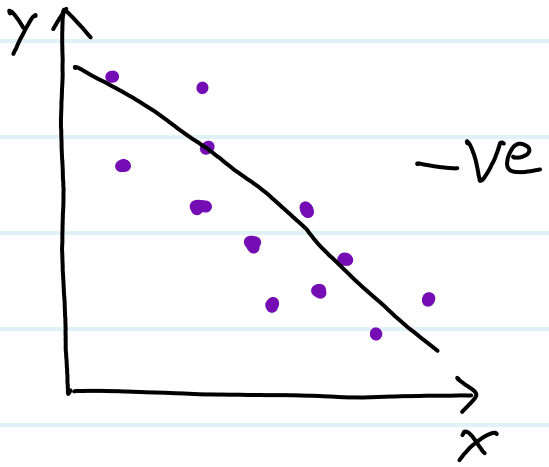
x_2 and y = 0.05

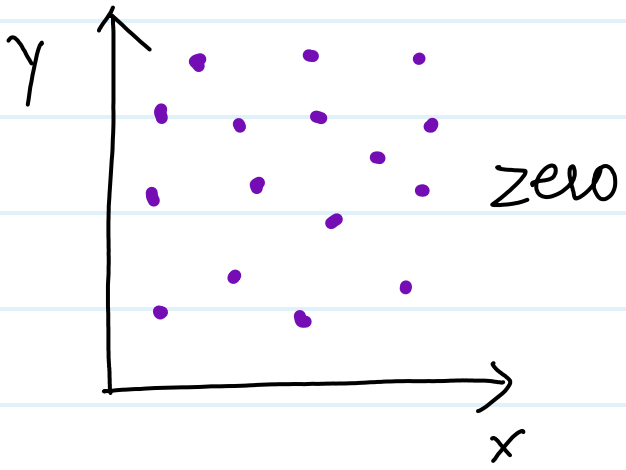
x_3 and y = −0.89

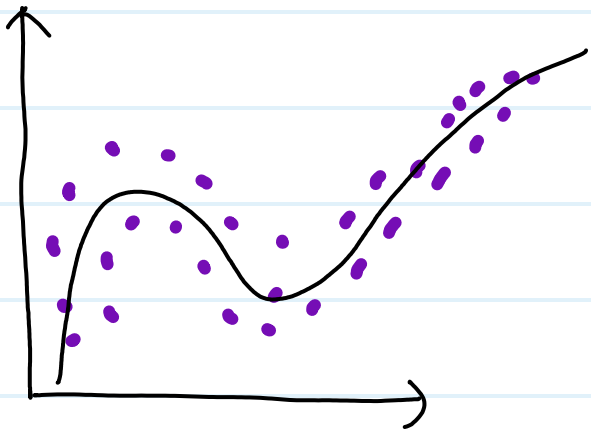
x_4 and y = −0.01

cons → It can't identify non-linear co-relation









II) Spearman Rank co-relation coeff.

$$\rho(X_R, Y_R) = 1 - \frac{6\Sigma d^2}{n(n^2-1)}$$

Range -1 to $+1$

X.	X_R ✓	Y	Y_R ✓	d	d^2	Σd^2
2	4	10	5	-1	1	26
5	1	80	3	-2	4	
3	3	40	2	1	1	
1	5	50	1	4	16	
4	2	20	4	-2	4	

$$\Rightarrow 1 - \frac{6 \times 26}{5(5^2-1)}$$

$$\Rightarrow -0.3$$

variables

- Quanti.
  - ① Descret
  - ② Continous
- Qualit.
  - ① Nominal
  - ② Ordinal

X_1	X_2	X_3	X_4	X_5
2	1025	A	T	10k
3	11.11	B	F	20k
1	21.05	B	T	50k
4	4.05	D	T	60k
0		C	F	

variabe $X = 4$

$X = [3,4,2,1,5]$

Random variable

$X \approx 5$

$X_{(H,T)}$ = Coin

X_1	X_2	X_3	X_4	Y
2	10 15	A	T	-
4	20 25	B	F	-
6	10.05	C	T	-

① Descret

[whole no. cat.]

② Continous

[Decimal]

X_1 X_2 X_3

Feature
OR
Independent variable
OR
metrics

Y

Target
OR
Dependent variable
OR
Lable